

Algebraic & Geometric Methods in Statistics

Outline and some illustrative examples in nonlinear statistics

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Created for Math/Stat 561

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Lecture plan from Monday

We will continue now with the following topics:

- Probability Primer (Chapter 2) and
- Conditional Independence (Chapter 4)

Some probability and a bit of stats

Topics:

- discrete probability distributions: basic notation.
- The model of independence.
- Limiting distributions - see section 2.3 and 2.5 in the book.

Lecture on board.

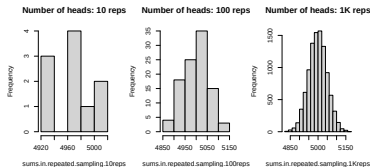
Why do we care about distributions?

Who cares about model fitting and testing whether we have the correct model in the first place?

Why do I have to understand a model?

Simulation of a coin toss

- Let x be the random variable recording the outcome of a coin toss:
 - $x_i = 0$ if we see Tail on the i -th trial (toss),
 - $x_i = 1$ if we see heads on the i -th trial.
- Fix $n = 10000$.
- $Y =$ the number of heads.
 - Is the number of heads supposed to be $n/2$? How far off is it? Does it vary? What does this mean?



- Sampling distribution of Y appears to have a mean around *the expected number of heads when a fair coin is tossed*, which is about $n/2$.
- The more times we repeat the experiment of n coin tosses, the closer Y gets to its expected value – this can be measured by looking at both the mean and the variance of Y .

Means of Y

4969.500

5000.300

4999.317

Vars of Y

3264.131

3264.131

2493.795

Question:

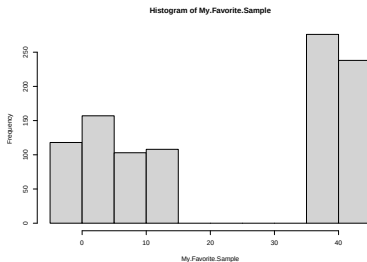
is it possible that something similar to this always happens?

- As we will see, the sampling distribution of Y is approximately *normal* with mean equal to the expected value of X .
- In other words, the example above illustrates a known result—the **Central Limit Theorem**, one of the cornerstone results used in inference.
- You should already be familiar with it from your probability class.

Importance of sampling distributions

- Sampling distributions tell a story about the model behind the data (i.e., the probability distribution or population from which the data was sampled);
- they give a glimpse into how it was generated.

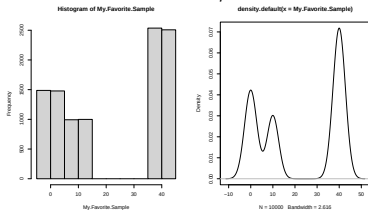
Example



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-2.809	1.354	38.053	22.655	39.941	43.481

Hmm...

- Is it strange to see “two bumps” in the histogram instead of one, as usual?
- Maybe the sample size is too small, we need to simulate more data?

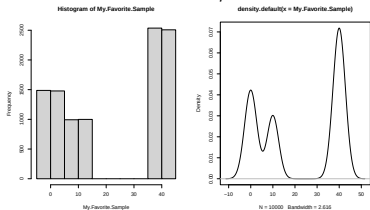


AhaMoment!

What do you see?

Hmm...

- Is it strange to see “two bumps” in the histogram instead of one, as usual?
- Maybe the sample size is too small, we need to simulate more data?



AhaMoment!

What do you see?

- This data is *not* being drawn from anything like a normal distribution.
- Consequently, knowing simply the mean and the variance ... is not enough to understand the data, that is, the data-generating mechanism behind it.

... Wait, what was that?!

This was an example of a **mixture** of normal distributions.

Check out the handout on Campuswire to see more, and ponder some important questions.

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